DETECTION AND AUTOMATED MITIGATION OF APPLICATION-LAYER DDOS ATTACKS IN CONTAINERIZED ENVIRONMENTS

SE-401 • *Attack Focus: Slowloris*

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2 Problem Description & Motivation

Attack Focus: Slowloris — a slow-rate HTTP attack that opens many connections and sends partial requests to keep them alive, exhausting the server's connection and thread pool without using high bandwidth.

The Problem

Slowloris sends seemingly legitimate but incomplete HTTP requests. Each connection is kept alive with periodic partial headers, tying up an application thread or connection slot without ever completing the request.



Many Slow Connections → Threads Exhausted → New Users Blocked → Service Unavailable

Main Goal

Detect low-and-slow Slowloris-style connection behaviour and automatically isolate the offending container, while keeping the service available to legitimate users.

Why it's Important?

- Slowloris traffic looks like normal, slow clients — no malformed packets or huge volume.
- Traditional volume-based detection misses it: total bandwidth stays very low.
- Held-open connections deplete the thread and connection pool, blocking real users.
- Because it acts slowly, automated real-time response is needed — manual reaction is too late.

RELEVANCE TO CONTAINERIZED ENVIRONMENTS

How Slowloris specifically threatens multi-tenant containers

Default deployment model

Multi-service, multi-tenant containers on shared infrastructure are now the norm — Docker and Kubernetes power most modern app deployments.

Small per-container thread pools

Containers are given small connection and worker-thread limits, so a Slowloris attack that would be minor on a large VM can exhaust a single container's pool very quickly.

Static defenses, static config

Existing container security (namespaces, cgroups, network policies) is configured in advance — it doesn't react to a slow-rate connection attack in progress.

Real, growing attack surface

A single Slowloris-exhausted container can become a launch point for attacking neighbouring services on the same host or cluster.

PROPOSED DETECTION & MITIGATION APPROACH

Tuned specifically to catch Slowloris's low-and-slow signature

1



Monitor

Sniff live connection state to the containerized service in real time — duration, bytes/sec and header-completion rate per connection.

2



Classify

XGBoost model and adaptive threshold flag Slowloris-style flows: long-lived, low-throughput connections with incomplete headers.

3



Isolate

Runtime API pauses or disconnects the offending container's slow connections immediately.

4



Restore

Isolation auto-lifts after a cooldown if connection behaviour returns to normal.

Contribution: *not detection accuracy alone — the automated, container-native isolation loop that acts on Slowloris's slow-rate signature in real time.*

INITIAL RESEARCH REFERENCES

1. Deressu et al., 2026

“A Survey of Application Layer Distributed Denial of Service Detection: Approaches, Models, and Datasets” — Artificial Intelligence Review.

2. Sharif et al., 2024

“Detection of application-layer DDoS attacks using machine learning and genetic algorithms” — Computers & Security.

3. Praseed & Thilagam, 2022

“HTTP request pattern based signatures for early application layer DDoS detection” — Journal of Information Security and Applications.

4. Cambiaso et al.

Slow DoS attack taxonomy and analysis, including Slowloris connection-exhaustion behaviour.

5. OWASP

Denial of Service Cheat Sheet — application-layer DoS and Slowloris mitigation concepts.

6. Kubernetes Documentation

Resource Management, Network Policies and Security Controls.

THANK YOU